

9 Each employee in a finance company has a computer and an electronic account.

- (a) Each computer has an operating system. One function of the operating system is to manage memory.

Describe the role of the operating system in managing memory.

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- (b) The finance company is concerned that a hacker may perform a brute-force attack to gain access to an employee's account.

- (i) Describe how the hacker would perform the brute-force attack.

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- (ii) Give **three** cyber security solutions that could be used to help prevent a brute-force attack from being successful.

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- (c) The company uses a digital currency to purchase a new car.

- (i) State what is meant by a digital currency.

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- (ii) The digital currency uses a process that involves a digital ledger. The ledger has time stamped transactions that cannot be altered.

Give the name of this process.

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